

Xeno-estrogenic compounds in precipitation.

The exposure to some chemicals can lead to hormone disrupting effects. Presently, much attention is focused on so-called xeno-estrogens, synthetic compounds that interact with hormone receptors causing a number of reactions that eventually lead to effects related to reproduction and development. The current study was initiated to investigate the presence of a number of such compounds in precipitation as a follow-up on a previous study in which pesticide concentrations in air and precipitation were determined. Rainwater samples were collected at about 50 locations in The Netherlands in a four week period. The samples were analysed for bisphenol-A, alkylphenols and alkylphenol ethoxylates, phthalates, flame retardants and synthetic musk compounds. The results clearly indicated the presence of these compounds in precipitation. The concentrations ranged from the low ng l(-1) range for flame retardants to several thousands of ng l(-1) for the phthalates. Bisphenol-A was found in 30% of the samples in concentrations up to 130 ng l(-1), while alkylphenols and alkylphenol ethoxylates were found in virtually all locations in concentrations up to 920 ng l(-1) for the individual compounds. Phthalates were by far the most abundant xeno-estrogens in the precipitation samples and were found in every sample. Di-isodecyl phthalate was found in a surprisingly high concentration of almost 100 000 ng l(-1). Polybrominated flame retardants were found in the low ng l(-1) range and generally in less than 20% of the samples. Noticeable was the finding of hexabromocyclododecane, a replacement for the polybrominated diphenyl ethers at one location in a concentration of almost 2000 ng l(-1). Finally, as expected, synthetic musk compounds were detected in almost all samples. This is especially true for the polycyclic musks HHCB and AHTN. Nitro musks were found, but only on a few locations. Kriging techniques were used to calculate precipitation concentrations in between actual sampling locations to produce contour plots for a number of compounds. These plots clearly show located emission sources for a number of compounds such as bisphenol-A, nonylphenol ethoxylate, phthalates and AHTN. On the contrary, the results for HHCB and some phthalates indicated diffuse emission patterns, probably as the result of the use of consumer products containing these compounds.